

Statistical Modelling and Inference

Assignment 4

Spring 2 2022

ASSIGNMENT CHECKLIST

- Have you shown all of your working, including probability notation where necessary?
- Have you included all R output and plots to support your answers where necessary?
- Have you included all of your R code?
- Have you made sure that all plots and tables each have a caption?
- Is the submission a single pdf file - correctly orientated, easy to read? If not, penalties apply.
- Assignments emailed instead of submitted by the online submission on BB will not be marked and will receive zero.
- Q1-Q3 may be typed or hand written and scanned in as a pdf

Q1

This questions may be typed or hand written and scanned in as a pdf.

The aim of this question is to familiarise your with linearising models and dealing with transformations in linear regression.

The saturated growth equation is useful for modelling the growth of some animal species. It can be written as

$$Y = \frac{\alpha x}{\delta + x},$$

where Y is some physical measurements of animal size, x represents time, and α and δ are unknown parameters.

- (a) Linearise the saturated growth equation. [2 marks]
- (b) Find $\hat{\alpha}$ and $\hat{\delta}$ based on the least-squares estimate of the parameters of the linearised model. [2 marks]
- (c) Explain why the approach in part (b) is different to fitting the saturated model *directly* using the method of least squares. **Hint:** What is the function we are minimizing in each case? [3 marks]
- (d) A study was conducted in 2002 to investigate the growth of harbour seals. Carcasses of seals that drifted ashore were collected. Their length (in cm) were measured and their age (in years) were estimated based on teeth. The linearised saturated growth model was fitted to the data using R and the following output was obtained. Note that some entries in the outputs in this question are missing and you are to calculate this. The suffix `.trans` is used to indicate that a transformation was applied to the variable.

```
seal.lm <- lm(Length.trans ~ Age.trans)
summary(seal.lm)
```

```
##
## Call:
```

```
## lm(formula = Length.trans ~ Age.trans)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.986e-04 -1.731e-04 -2.973e-05  1.127e-04  5.092e-04
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.075e-03  9.825e-05  61.830  < 2e-16 ***
## Age.trans    ?????????  2.897e-04   8.423  1.27e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0002495 on 13 degrees of freedom
## Multiple R-squared:  0.8451, Adjusted R-squared:  0.8332
## F-statistic: 70.94 on 1 and 13 DF,  p-value: 1.267e-06
```

i) How many observations are in the data?

[1 mark]

ii) A few observations from the data are given below. For the following observations, write down the response vector $\mathbf{y}$ and the design matrix $\mathbf{X}$ for the linearised model.

[2 marks]

Length	Age
489	4
476	3
513	7
382	1

iii) Using Part (b) and the R output above, calculate $\hat{\alpha}$ and $\hat{\delta}$ from the saturated growth curve for this data, and give the best fitting saturated growth curve.

[6 marks]

iv) Find the 90% confidence interval for the intercept term in the linearised model.

[4 marks]

v) The following output gives the 90% prediction interval for `Length.trans` of a seal aged 6 years, based on the **linearised model**. Find the corresponding 90% prediction interval for the **length of a seal** aged 6 years.

[5 marks]

```
x0 <- tibble(Age.trans=??)
predict(seal.lm, newdata = x0, interval = "prediction", level = 0.9)
```

```
##           fit           lwr           upr
## 1 0.006481724 0.006023172 ????????????
```

[Question total: 25]

Solutions:

(a)

$$\begin{aligned} Y &= \frac{\alpha x}{\delta + x} \\ \frac{1}{Y} &= \frac{\delta + x}{\alpha x} \\ &= \left(\frac{\delta}{\alpha}\right) \left(\frac{1}{x}\right) + \left(\frac{1}{\alpha}\right) \\ Y^* &= \beta_1 x^* + \beta_0 \end{aligned}$$

(b) Let $\hat{\beta}_0$ and $\hat{\beta}_1$ denote the least squares estimate of β_0 and β_1 respectively. From Part (a), we have that $\frac{1}{\hat{\alpha}} = \hat{\beta}_0$ and $\frac{\hat{\delta}}{\hat{\alpha}} = \hat{\beta}_1$. Hence,

$$\hat{\alpha} = \frac{1}{\hat{\beta}_0} \quad \text{and} \quad \hat{\delta} = \hat{\alpha} \hat{\beta}_1 = \frac{\hat{\beta}_1}{\hat{\beta}_0}.$$

(c) With linearization, least square estimation is minimizing

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n (y_i^* - \beta_1 x_i^* - \beta_0)^2 = \sum_{i=1}^n \left(\frac{1}{y_i} - \frac{1}{\alpha} - \frac{\delta}{\alpha} \frac{1}{x_i} \right)^2.$$

Whereas without linearization, we are directly minimizing

$$Q(\alpha, \delta) = \sum_{i=1}^n \left(y_i - \frac{\alpha x_i}{\delta + x_i} \right)^2.$$

These two functions are obviously different. The error structure has also changed after linearization.

(d) i) From the output, we know that $n - p = 13$ and $p - 1 = 1$. Hence, $n = 15$.

ii)

$$\mathbf{y}^* = \begin{bmatrix} \frac{1}{489} \\ \frac{1}{476} \\ \frac{1}{513} \\ \frac{1}{382} \end{bmatrix}, \quad \mathbf{X}^* = \begin{bmatrix} 1 & \frac{1}{4} \\ 1 & \frac{1}{3} \\ 1 & \frac{1}{7} \\ 1 & 1 \end{bmatrix}$$

iii) From the R output, we have $\hat{\beta}_0 = 6.075 \times 10^{-3}$ and $\hat{\beta}_1 = 8.423 \times (2.897 \times 10^{-4}) = 2.44 \times 10^{-3}$.

Hence, $\hat{\alpha} = \frac{1}{6.075 \times 10^{-3}} = 164.6091$ and $\hat{\delta} = \frac{2.44 \times 10^{-3}}{6.075 \times 10^{-3}} \approx 0.4017$. The fitted model is

$$Y = \frac{164.6091x}{0.4017 + x}.$$

iv) The critical value in this case is $t_{13,0.05} \approx 1.7709$ (using `qt(0.95, 13)` in R). From the R output, we have $\hat{\beta}_0 = 6.075 \times 10^{-3}$ and $SE(\hat{\beta}_0) = 9.825 \times 10^{-5}$. It follows that

$$CI = \hat{\beta}_0 \pm t_{13,0.05} SE(\hat{\beta}_0) = 6.075 \times 10^{-3} \pm 1.7709 (9.825 \times 10^{-5}) \approx (0.0059, 0.0062).$$

v) The upper limit of the CI for `Length.trans` can be deduced using

$$\text{upp} = \text{fit} + (\text{fit} - \text{lwr}) = 0.006481724 + (0.006481724 - 0.006023172) = 0.006940276$$

We can back transform the PI for `Length.trans` to obtain an approximate PI for `Length`:

$$PI_{\text{Length}} = \left(\frac{1}{\text{upr}}, \frac{1}{\text{lwr}} \right) = \left(\frac{1}{0.006940276}, \frac{1}{0.006023172} \right) = (144.0865, 166.0255)$$

Q2

This questions may be typed or hand written and scanned in as a pdf.

A survey, conducted in a city by a local Internet provider, collected data for the following variables from 35 randomly selected households.

- Y : number of hours of Internet used in a week
- X_1 : number of children in the household
- X_2 : net household income (in dollars) in the previous week
- X_3 : number of years of formal education of the highest income earner in the household

The values for the first 4 households are shown below.

Household	1	2	3	4
Y	12.8	9.4	14	15.6
X_1	1.0	0.0	2	3.0
X_2	1590.0	968.0	732	780.0
X_3	15.0	11.0	12	13.0

Consider the multiple regression model

$$M: \mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $Y_i \sim N(\eta_i, \sigma^2)$ independently for $i = 1, 2, \dots, n$ and $\boldsymbol{\eta} = X\boldsymbol{\beta}$.

- (a) Write down the dimensions of $\mathbf{Y}$, X , $\boldsymbol{\beta}$, and $\boldsymbol{\epsilon}$.

[4 marks]

- (b) Write down $\boldsymbol{\beta}$ in full, and the first four rows of X and $\mathbf{y}$ (the vector of observed values).

[3 marks]

[Question total: 7]

Solutions:

- (a) There are $n = 35$ subjects (households) and $p = 4$ parameters $(\beta_0, \beta_1, \beta_2, \beta_3)$.

- $\mathbf{Y}$ is 35×1 ,
- $\boldsymbol{\epsilon}$ is 35×1 ,
- $\boldsymbol{\beta}$ is 4×1 , and
- X is 35×4 .

- (b) We have

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 1 & 1590 & 15 \\ 1 & 0 & 968 & 11 \\ 1 & 2 & 732 & 12 \\ 1 & 3 & 780 & 13 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 12.8 \\ 9.4 \\ 14.0 \\ 15.6 \\ \vdots \end{pmatrix}.$$

Q3

This questions may be typed or hand written and scanned in as a pdf.

Suppose $y_1, y_2, \dots, y_n$ are independent exponential observations with probability density function

$$f(y; \theta) = \frac{1}{\theta} e^{-y/\theta} \quad \text{for } y \geq 0, \quad \theta > 0.$$

(a) Write down the log-likelihood $\ell(\theta; \mathbf{y})$.

[2 marks]

(b) Hence, find the maximum likelihood estimate $\hat{\theta}$.

[2 marks]

(c) Find the Fisher information I_{θ} .

[3 marks]

[Question total: 7]

Solutions:

(a) The likelihood function is

$$\begin{aligned} L(\theta, \mathbf{y}) &= \prod_{i=1}^n \frac{1}{\theta} e^{-y_i/\theta} \\ &= \frac{1}{\theta^n} e^{-\sum y_i/\theta}. \end{aligned}$$

Hence the log-likelihood is

$$\ell(\theta, \mathbf{y}) = \log(L(\theta, \mathbf{y})) = -n \log \theta - \frac{\sum y_i}{\theta}.$$

(b) The score function is

$$S(\theta, \mathbf{y}) = \frac{\partial \ell}{\partial \theta} = \frac{-n}{\theta} + \frac{\sum y_i}{\theta^2}.$$

Setting $S(\theta, \mathbf{y})$ equal to zero and solving gives the MLE

$$\hat{\theta} = \frac{\sum y_i}{n} = \bar{y}.$$

(c) The Fisher information defined as

$$I_{\theta} = E \left(-\frac{\partial^2 \ell}{\partial \theta^2} \right).$$

The second derivative of the log-likelihood is

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2 \sum y_i}{\theta^3}.$$

Therefore,

$$\begin{aligned} E \left(-\frac{\partial^2 \ell}{\partial \theta^2} \right) &= 2 \sum_{i=1}^n \frac{E(Y_i)}{\theta^3} - \frac{n}{\theta^2} \\ &= 2 \frac{n\theta}{\theta^3} - \frac{n}{\theta^2} \\ &= \frac{n}{\theta^2}. \end{aligned}$$

Q4

The aim of this question is to explore the concepts on ANCOVA and the three types of regression models associated to it: identical, parallel, and separate. You will also get to whet your skills on performing some analysis on contrasts.

An agriculture company has developed three new variants of fast growing crop, and wishes to investigate their sensitivity to soil salinity. Seeds of each plant type were planted in different sites with varying levels of

soil salinity (measured in decisiemens per meter (dS/m)). The height (in 10cm) of the crop was measured at 15 days after they were planted. The data is given in `crop.csv`, available from BB. It contains the following variables for 85 crops:

Variable	Details
Crop_no	Unique identifier for each crop
Type	Variant of crop (Type I, II, or III)
Salinity	Salinity level of the soil (dS/m)
Height	Height of the crop (10cm)

- (a) Produce a scatterplot of Height against Salinity level by crop type. Describe the relationship.

[3 marks]

- (b) Fit the following linear models in R:

- Model 1 – identical regression lines: Height on Salinity level
- Model 2 – parallel regression lines: Height on Type and Salinity level
- Model 3 – separate regression lines: Height on Type and Salinity level with interaction

Consider Model 3 (separate regression lines). Write down the line of best fit for the relationship between Height on Type and Salinity level for each of Type I, Type II, and Type III crop separately.

[3 marks]

- (c) Test for a statistically significant interaction term in Model 3 (separate regression lines) at the 5% significance level. Clearly state the null and alternative hypotheses, the test statistic, the P-value, and your conclusion.

[2 marks]

- (d) Calculate the BIC for all three models. Which model has the best fit according to BIC?

[2 marks]

- (e) Assess the assumptions for Model 3 (separate regression lines) using appropriate diagnostic plots.

[4 marks]

- (f) The company is interested in the difference between crop types II and III. Based on Model 3 (separate regression lines), calculate the estimated difference in mean height between Type II and Type III crops (that is, $\hat{\mu}_{\text{Type II}} - \hat{\mu}_{\text{Type III}}$). Then calculate the corresponding 95% confidence interval. [**Hint: Review the lecture on contrasts.**]

[3 marks]

[Question total: 17]

Solutions:

- (a) See the code and output attached at the end for the scatterplot.

There is a weak, negative, and non-linear relationship between Height against Salinity level. The lines for different types of crops appear to have different slopes and intercepts.

- (b) See the output for separate regression.

For type I crop:

$$\text{Height} = 5.7874 - 0.2902 \times \text{Salinity}$$

For type II crop:

$$\begin{aligned} \text{Height} &= (5.7874 + 2.9467) + (-0.2902 - 1.6772) \times \text{Salinity} \\ &= 8.7341 - 1.9674 \times \text{Salinity} \end{aligned}$$

For type III crop:

$$\begin{aligned}\text{Height} &= (5.7874 + 22.4455) + (-0.2902 - 7.9971) \times \text{Salinity} \\ &= 28.2329 - 8.2873 \times \text{Salinity}\end{aligned}$$

(c) The full model (i.e., Model 3, separate regression lines) is

$$\text{Height}_i = \beta_0 + \beta_1 \times \text{Salinity}_i + \beta_2 \times \text{TypeII} + \beta_3 \times \text{TypeIII} + \beta_4 \times \text{Salinity}_i \times \text{TypeII} + \beta_5 \times \text{Salinity}_i \times \text{TypeIII} + \epsilon_i$$

We are testing $H_0 : \beta_4 = \beta_5 = 0$ vs $H_1 : \text{at least one of } \beta_4 \text{ and } \beta_5 \text{ is not zero}$, where β_4 and β_5 are the coefficients of interaction terms in Model 3.

See the output for `anova(crop\_seperate, crop\_parallel)`.

The test statistic is $F = 24.234$ and the associated P-value is 6.2×10^{-9} . Based on this, there is sufficient evidence to reject H_0 at 5% significance level.

(d) From R, the BIC for the three models are as follows:

- Model 1 (identical regression lines): 448.3398
- Model 2 (parallel regression lines): 450.1701
- Model 3 (separate regression lines): 418.3894

Model 3 (separate regression lines) has the lowest BIC value and hence is preferred.

(e)

- **Linearity:** Residual versus Fitted (top left) shows the pattern is non-linear. So this assumption is not reasonable.
- **Homoscedascity:** Standardised residual versus Fitted (bottom left) shows unequal spread as move from left to right so this also not reasonable.
- **Normality:** Normal QQ-plot (top right) shows the points are reasonably close to the line, with deviation at the ends. The normality assumption is questionable.
- **Independence:** The height of one crop should not affect the height of the other crop so this is reasonable.

(f) The difference between the mean of type II and type III crops can be expressed as a linear combination of $\hat{\beta}$. Let $\mathbf{X}$ be the design matrix and $\beta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)^\top$ be the coefficients for the intercept, `Salinity`, `factor(Type)2`, `factor(Type)3`, `Height × factor(Type)2` and `Height × factor(Type)3`, respectively. Then setting

$$\lambda = (0, 0, 0, 1, -1, \text{Height}, -\text{Height})^\top$$

will give the difference in mean between Type II and Type II engine.

To obtain an estimate of this difference and the associated confidence interval (CI), we substitute `Height` with its estimated marginal mean (i.e., the mean of this covariate) and compute the CI using

$$CI = \lambda^\top \beta \pm t_{n-p, \alpha/2} se \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}.$$

This yields (see the R output)

$$\hat{\mu}_{\text{Type II}} - \hat{\mu}_{\text{Type III}} = -4.77$$

and

$$CI = (-6.56, -2.97).$$

Load in the data

```
# The here package can just make finding files nice, don't be put off by it.
crop <- read_csv(here::here("data/crop.csv"), col_types = cols())
head(crop) # Check it has read in properly
```

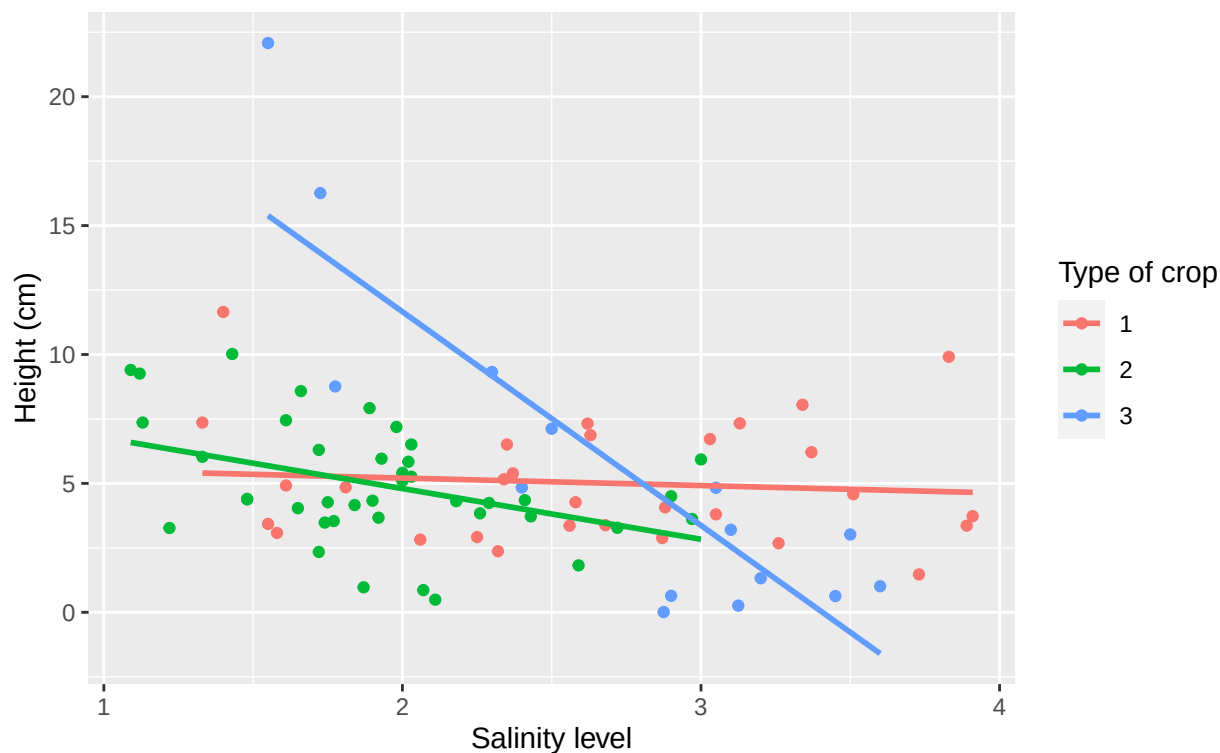
```
## # A tibble: 6 x 3
##   Height Salinity Type
##   <dbl>   <dbl> <dbl>
## 1    6.87    2.63     1
## 2    3.8     3.05     1
## 3    3.43    1.55     1
## 4    3.36    3.89     1
## 5    6.21    3.37     1
## 6    8.05    3.34     1
```

- Type has read in as a double, let's change that to a factor.

```
crop <- crop %>%
  mutate(Type = as.factor(Type))
```

Get a scatter plot

```
crop %>%
  ggplot(aes(x = Salinity, y = Height, colour = Type)) +
  geom_point() +
  geom_smooth(method = lm, formula = y ~ x, se = FALSE) +
  labs(y="Height (cm)", x="Salinity level", col="Type of crop")
```



Fit the 3 different ANCOVA models

```
identical <- lm(Height ~ Salinity, data = crop)
parallel <- lm(Height ~ Salinity + Type, data = crop)
separate <- lm(Height ~ Salinity * Type, data = crop)
```

```
summary(separate)
```

```
##
## Call:
## lm(formula = Height ~ Salinity * Type, data = crop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7630 -1.6844 -0.1351  1.7698  6.6924
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.7874     1.6536   3.500 0.000768 ***
## Salinity         -0.2902     0.5982  -0.485 0.628884
## Type2             2.9467     2.3131   1.274 0.206433
## Type3            22.4455     3.2230   6.964 8.75e-10 ***
## Salinity:Type2    -1.6772     1.0095  -1.661 0.100612
## Salinity:Type3    -7.9971     1.1517  -6.944 9.56e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.449 on 79 degrees of freedom
## Multiple R-squared:  0.4959, Adjusted R-squared:  0.464
## F-statistic: 15.54 on 5 and 79 DF,  p-value: 1.257e-10
```

Test for a significant interaction term

For this, we compare the parallel regressions to the separate regressions using an ANOVA:

```
anova(parallel, separate)
```

```
## Analysis of Variance Table
##
## Model 1: Height ~ Salinity + Type
## Model 2: Height ~ Salinity * Type
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      81 764.76
## 2      79 473.97  2    290.79 24.234 6.206e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

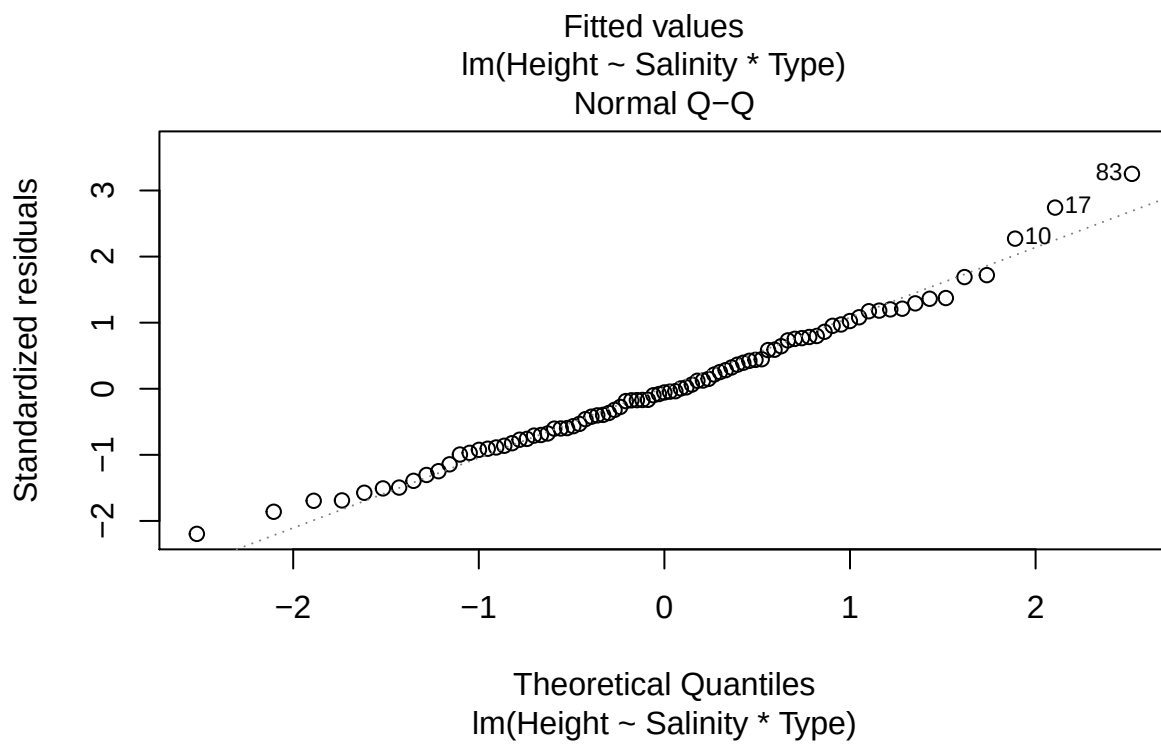
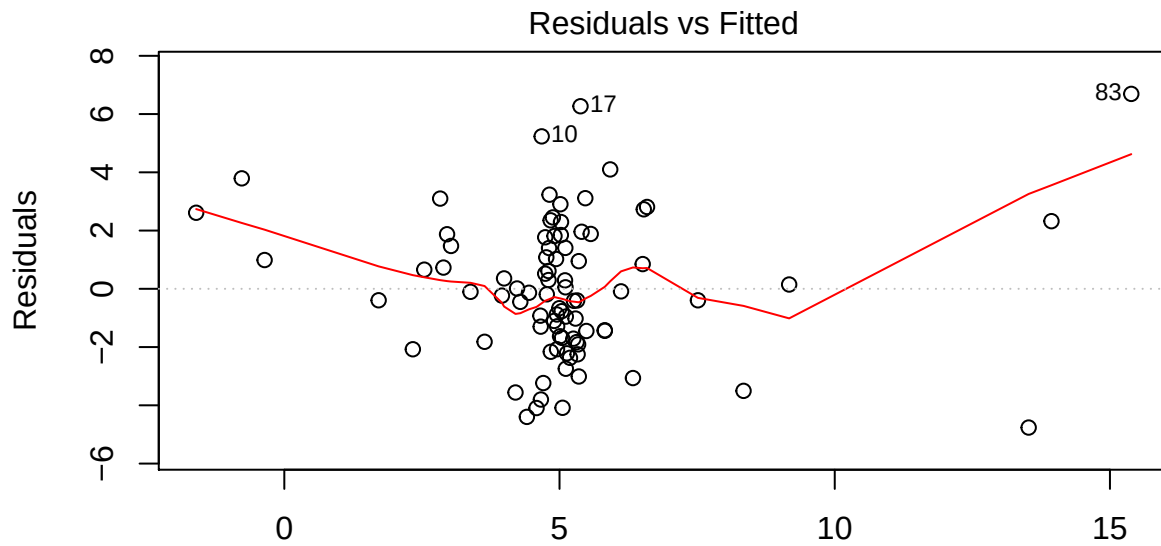
Obtain the BIC

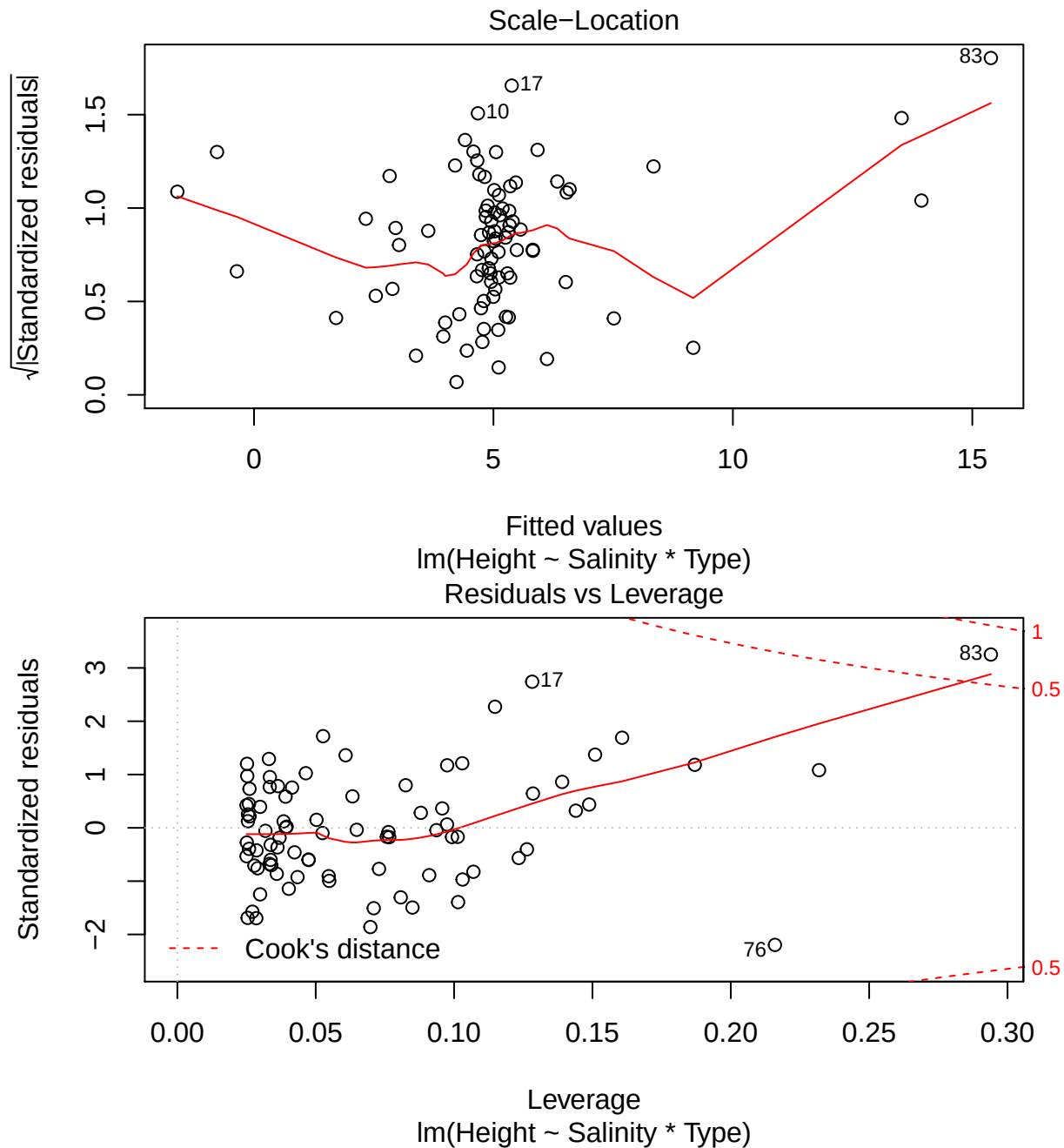
```
BIC(identical, parallel, separate)
```

```
##           df      BIC
## identical  3 448.3398
## parallel   5 450.1701
## separate   7 418.3894
```

Assumption plots:

```
plot(separate)
```





The hard one! contrast time

```
# Load the emmeans library
library(emmeans)
# create emmeans object, we are interested in breaking up the separate model
# by crop type.
crop_emmeans <- emmeans(separate, ~"Type")
```

NOTE: Results may be misleading due to involvement in interactions

```
# Create the contrast object
# This is saying we want to compare type II with type III
```

```
crop_contrast <- contrast(crop_emmeans,method=list("Type_II-III"=c(0,1,-1)))
crop_contrast
```

```
## contrast      estimate      SE df t.ratio p.value
## Type_II-III    -4.77 0.903 79  -5.280  <.0001
```

```
# get the CI
```

```
confint(crop_contrast)
```

```
## contrast      estimate      SE df lower.CL upper.CL
## Type_II-III    -4.77 0.903 79  -6.56    -2.97
##
## Confidence level used: 0.95
```

[[Assignment total: 56]]